Introduction

- The user interface part of a software product is responsible for all interactions with the end-user.
- An interface that is difficult to use leads to higher levels of user errors and ultimately leads to user dissatisfaction.
- When a user starts using a system, he builds a mental model of the system and expects the system behaviour to conform to it. For example, if a user action causes one type of system activity and response under some context, then the user would expect similar system activity and response to occur for similar user actions in similar contexts.
- Development of a good user interface usually takes significant portion of the total system development effort. For many interactive applications, as much as 50 per cent of the total development effort is spent on developing the user interface part.

Characteristics of a good user interface

Speed of learning

- A good user interface should be easy to learn.
- Speed of learning is hampered by complex syntax and semantics of the command issue procedures.
- A good user interface should not require its users to memorise commands.
- Neither should the user be asked to remember information from one screen to another while performing various tasks using the interface.
- Besides, the following three issues are crucial to enhance the speed of learning:
    - **Use of metaphors and intuitive command names**
        - The abstractions of real-life objects or concepts used in user interface design are called ***metaphors***.
        - **for example** - If the user interface of a text editor uses concepts similar to the tools used by a writer for text editing such as cutting lines and paragraphs and pasting it at other places, users can immediately relate to it.

        - Also, learning is facilitated by intuitive command names and symbolic command issue procedures.
    - **Consistency**
        - Once, a user learns about a command, he should be able to use the similar commands in different circumstances for carrying out similar actions.
        - This allows the user to extend his knowledge about one part of the interface to the other parts.
    - **Component-based interface**
        - Users can learn an interface faster if the interaction style of the interface is very similar to the interface of other applications with which the user is already familiar with.
        - This can be achieved if the interfaces of different applications are developed using some standard user interface components.

Speed of use

- Speed of use of a user interface is determined by the **time** and user **effort** necessary to initiate and execute different commands.
- The time and user effort necessary to initiate and execute different commands should be minimal.
- The most frequently used commands should have the smallest length or be available at the top of a menu to minimise the mouse movements necessary to issue commands.

Speed of recall

- Once users learn how to use an interface, the speed with which they can recall the command issue procedure should be maximised.
- Speed of recall is improved if the interface is based on some metaphors, symbolic command issue procedures, and intuitive command names.

Error prevention

- A good user interface should minimise the scope of committing errors while initiating different commands.
- Consistency of names, issue procedures, and behaviour of similar commands and the simplicity of the command issue procedures

minimise error possibilities. Also, the interface should prevent the user from entering wrong values.

Aesthetic and attractive

- A good user interface should be attractive to use.
- An attractive user interface catches user attention and fancy.

Feedback

- A good user interface must provide feedback to various user actions.
- Especially, if any user request takes more than few seconds to process, the user should be informed about the state of the processing of his request.

Support for multiple skill levels

- A good user interface should support multiple levels of sophistication of command issue procedure for different categories of users.
- Very cryptic and complex commands discourage a novice, whereas elaborate command sequences make the command issue procedure very slow and therefore put off experienced users.

Error recovery (undo facility)

- A good user interface should allow a user to undo a mistake committed by him while using the interface.
- If the users cannot recover even from very simple types of errors, they feel irritated, helpless, and out of control.

User guidance and on-line help

- Whenever users need guidance or seek help from the system, they should be provided with appropriate guidance and help.

Mode-based vs Modeless Interface

- A mode is a state or collection of states in which only a subset of all user interaction tasks can be performed.
- In a mode-based interface, different sets of commands can be invoked depending on the mode in which the system is, i.e., the mode at any instant is determined by the sequence of commands already issued by the user.

- In a modeless interface, the same set of commands can be invoked at any time during the running of the software.
- Thus, a modeless interface has only a single mode
- A mode-based interface can be represented using a state transition diagram, where each node of the state transition diagram would represent a mode. Each state of the state transition diagram can be annotated with the commands that are meaningful in that state.

Graphical User Interface (GUI) versus Text-based User Interface

- In a GUI multiple windows with different information can simultaneously be displayed on the user screen. This gives the user the flexibility to simultaneously interact with several related items at any time and can have access to different system information.
- Iconic information representation and symbolic information manipulation is possible in a GUI. Symbolic information manipulation such as dragging an icon representing a file to a trash for deleting.
- A GUI usually supports command selection using an attractive and user-friendly menu selection system.
- In a GUI, a pointing device such as a mouse or a light pen can be used for issuing commands. The use of which increases the efficacy of command issue procedure.
- On the flip side, a GUI requires special terminals with graphics capabilities for running and also requires special input devices such a mouse. On the other hand, a text-based user interface can be implemented even on a cheap alphanumeric display terminal.